

EXAMINATION OF STATISTICS DESCRIPTIVE

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Second Semester 2023-2024

BEM 1

NB: Presentation 1 Mark

✓ Exercise 1(7 mks)

The following distribution represents the repartition of salaries of workers in an enterprise in a ten of thousands.

Size of students in kg	Absolute frequency
10-50	38
50-90	41
90-130	19
130-170	28
170-210	18
210-250	29

Answer the following questions

1. What is the population study, its character and nature
2. Draw the histogram and the frequency polygon of this series
3. Calculate the modal class, arithmetic mean, median
4. Calculate the variance and coefficient of variation
5. Calculate the following quantities Q_2 , D_5 , and P_{50}

Exercise 2 (6 mks)

An industrial company in Cameroon manufacturing four products: plastic buckets, plastic chairs, plastic tables and cups, gives a rundown of their prices and quantities produced over two years 2023 base 2017 as on the table below

	2015		2023	
	P_{2017}	Q_{2017}	P_{2023}	Q_{2023}
Buckets	10	30	14	25
Chairs	20	25	26	25
Tables	50	15	60	30
Cups	25	32	50	28
Calculate	105	102	150	108

- a) Paasches price and quantity index
- b) Laspeyres price and quantity index

✓ Exercise 2(6 mks)

Consider the following distribution

X	22	25	10	31	16	14	13	22	19	20
Y	22	27	13	26	25	18	14	23	24	28

1. Calculate the persons correlation coefficient
2. Calculate the spearman correlation